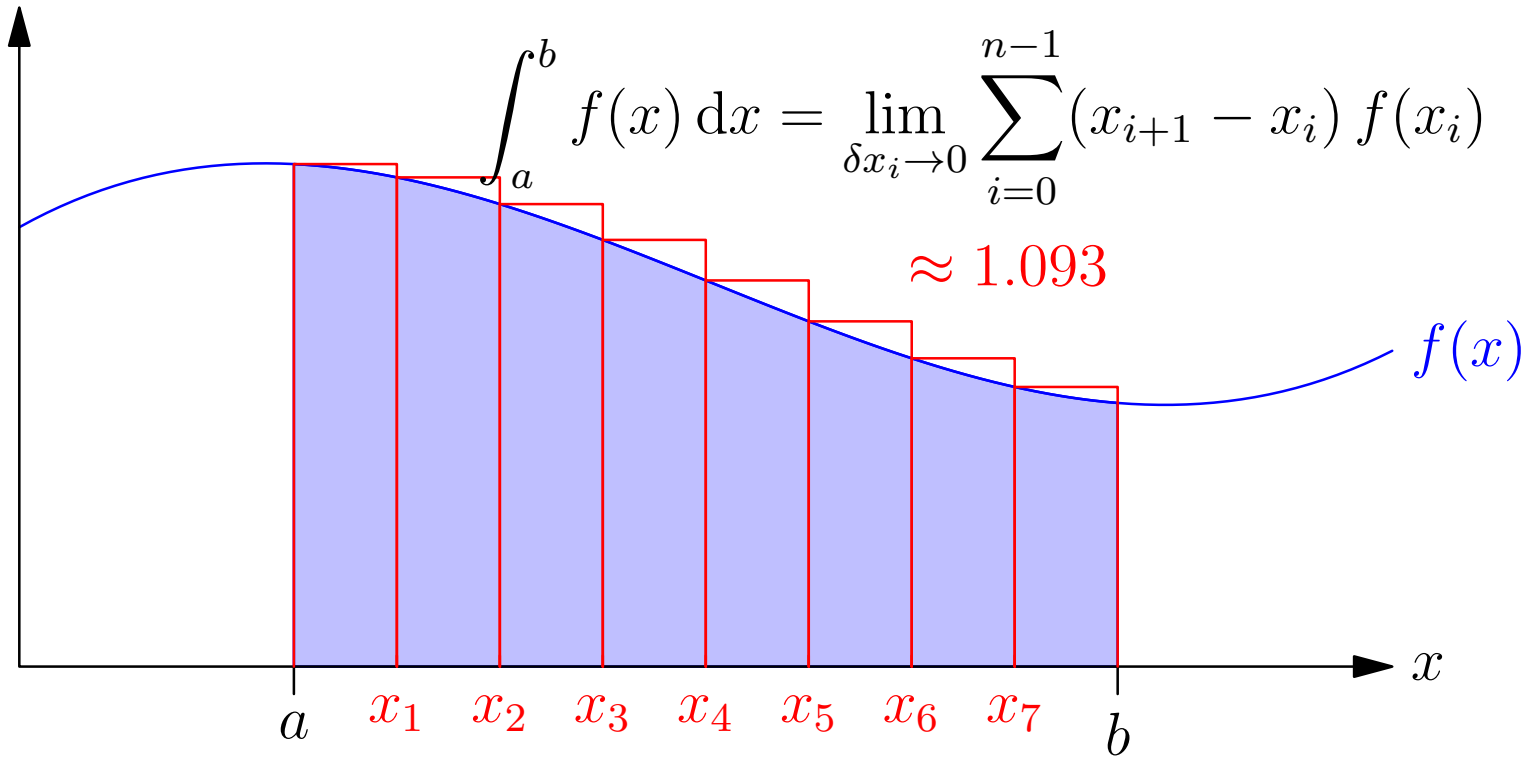
Mathematics of Machine Learning

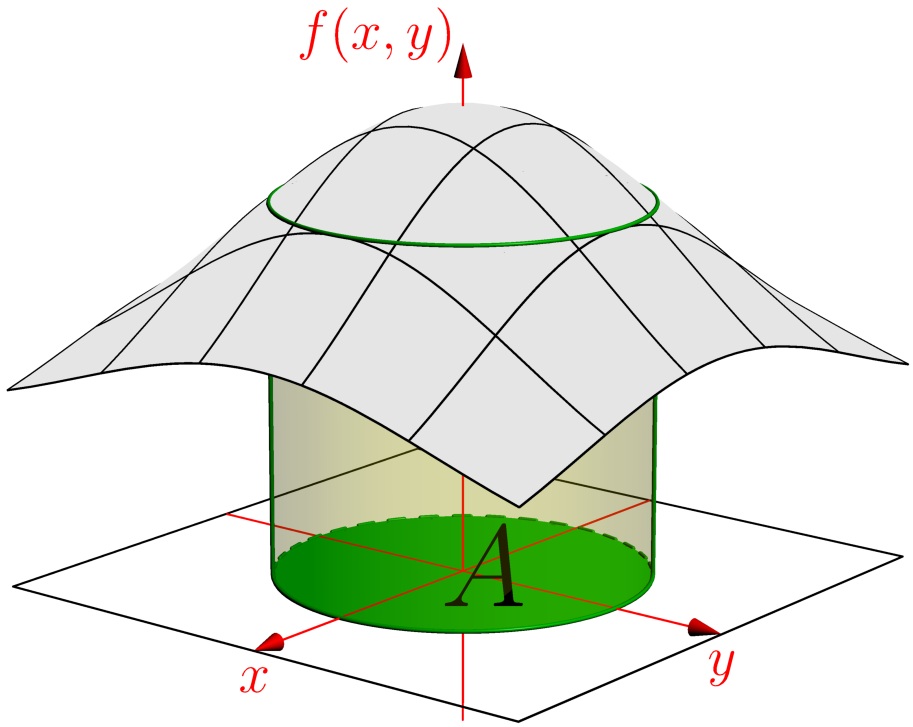
Integral Calculus



Riemann integration, integration by parts, substitution, change of variables

Outline

1. **Defining Integrals**
2. Doing Integrals
3. Changing Variables

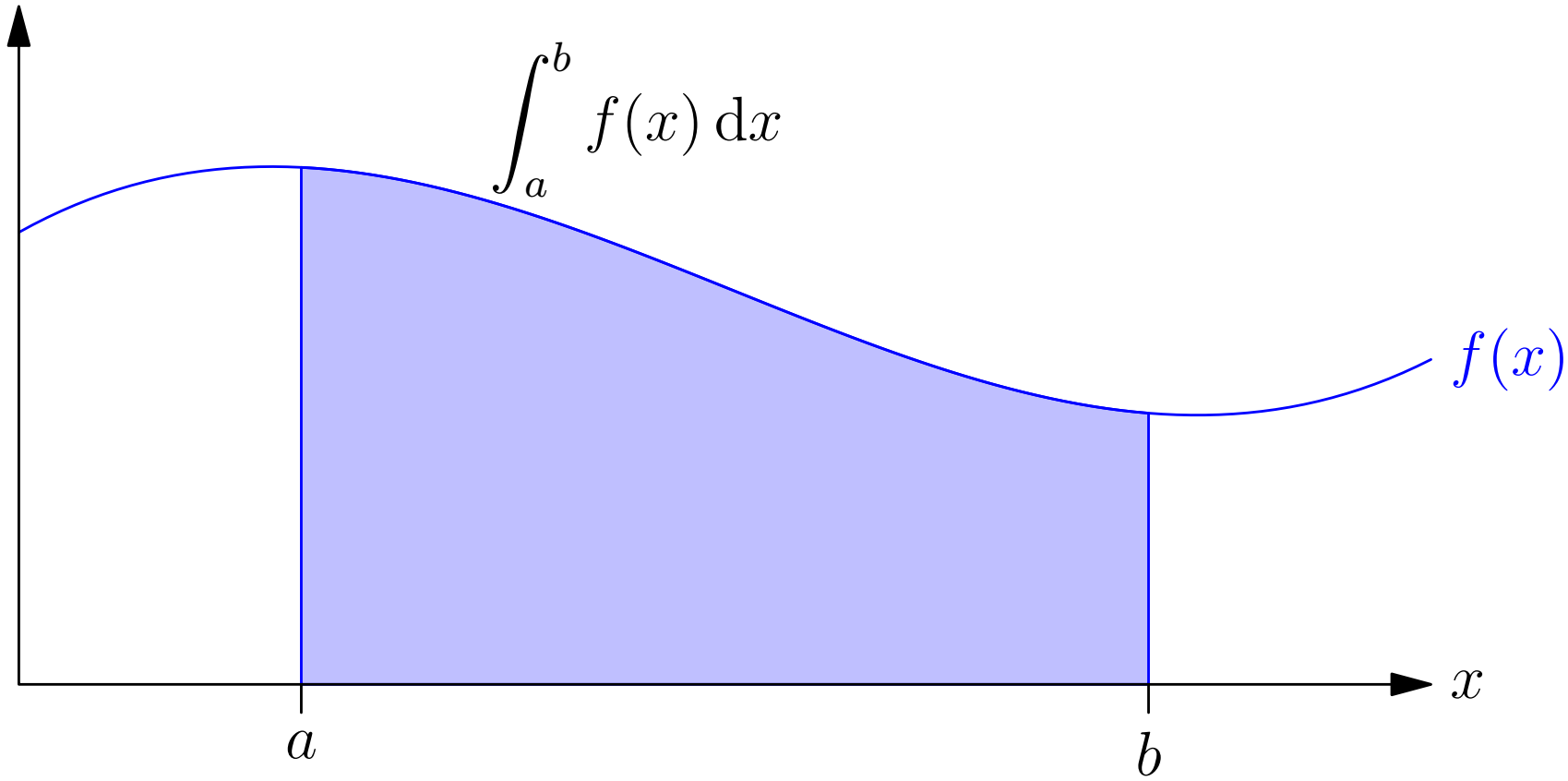


Riemann Integral

- Integrals represent area beneath a curve

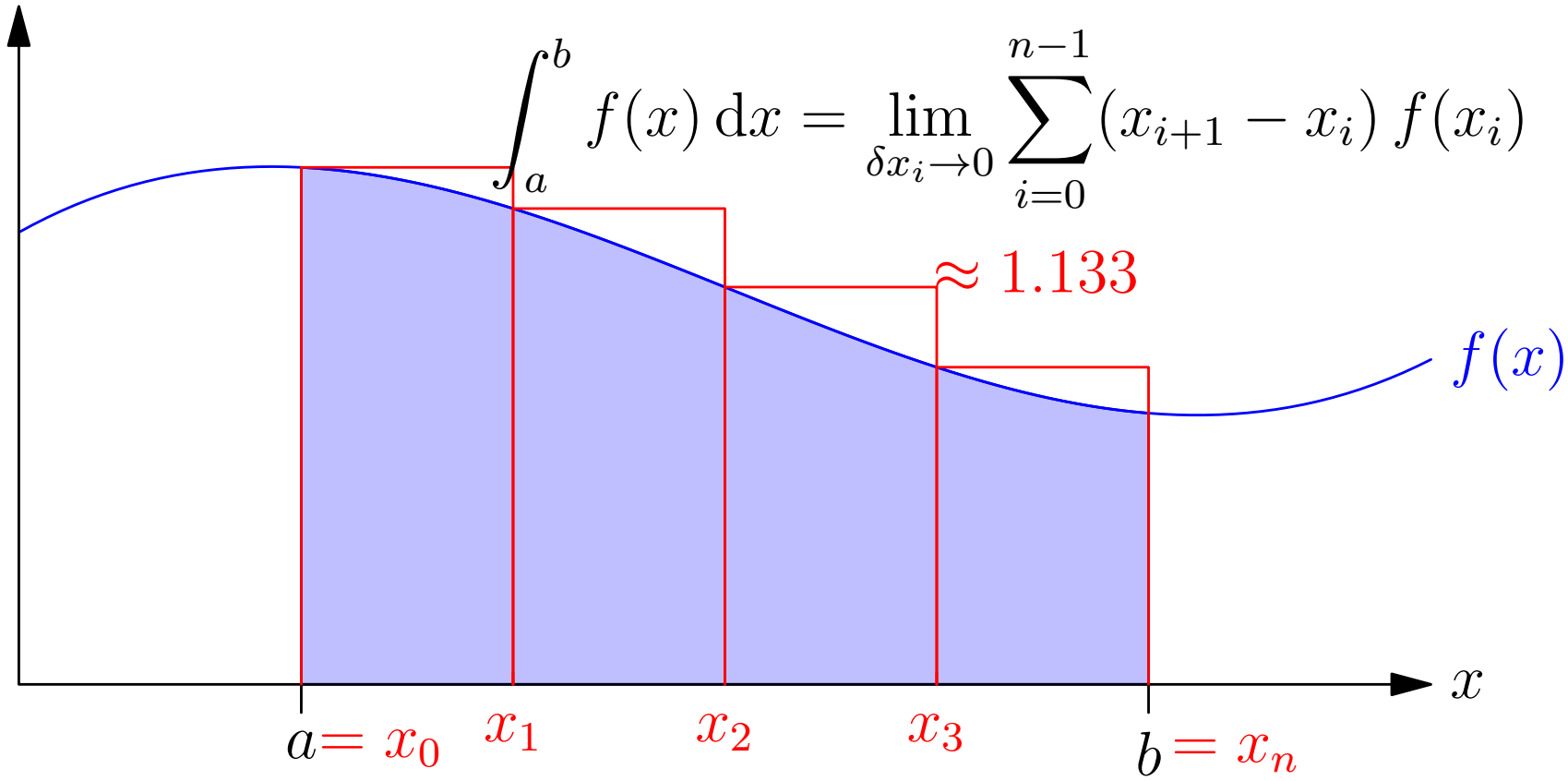
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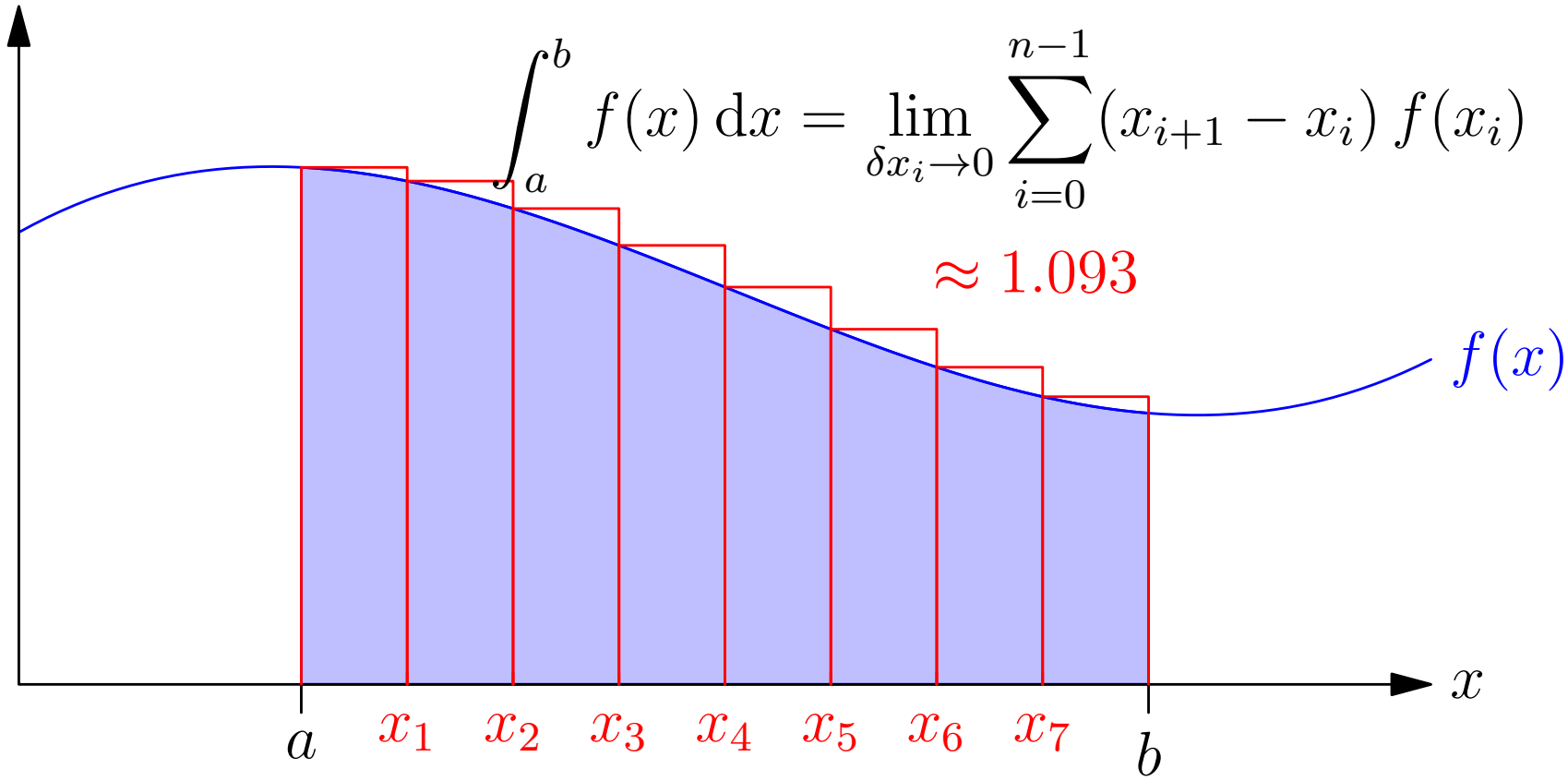
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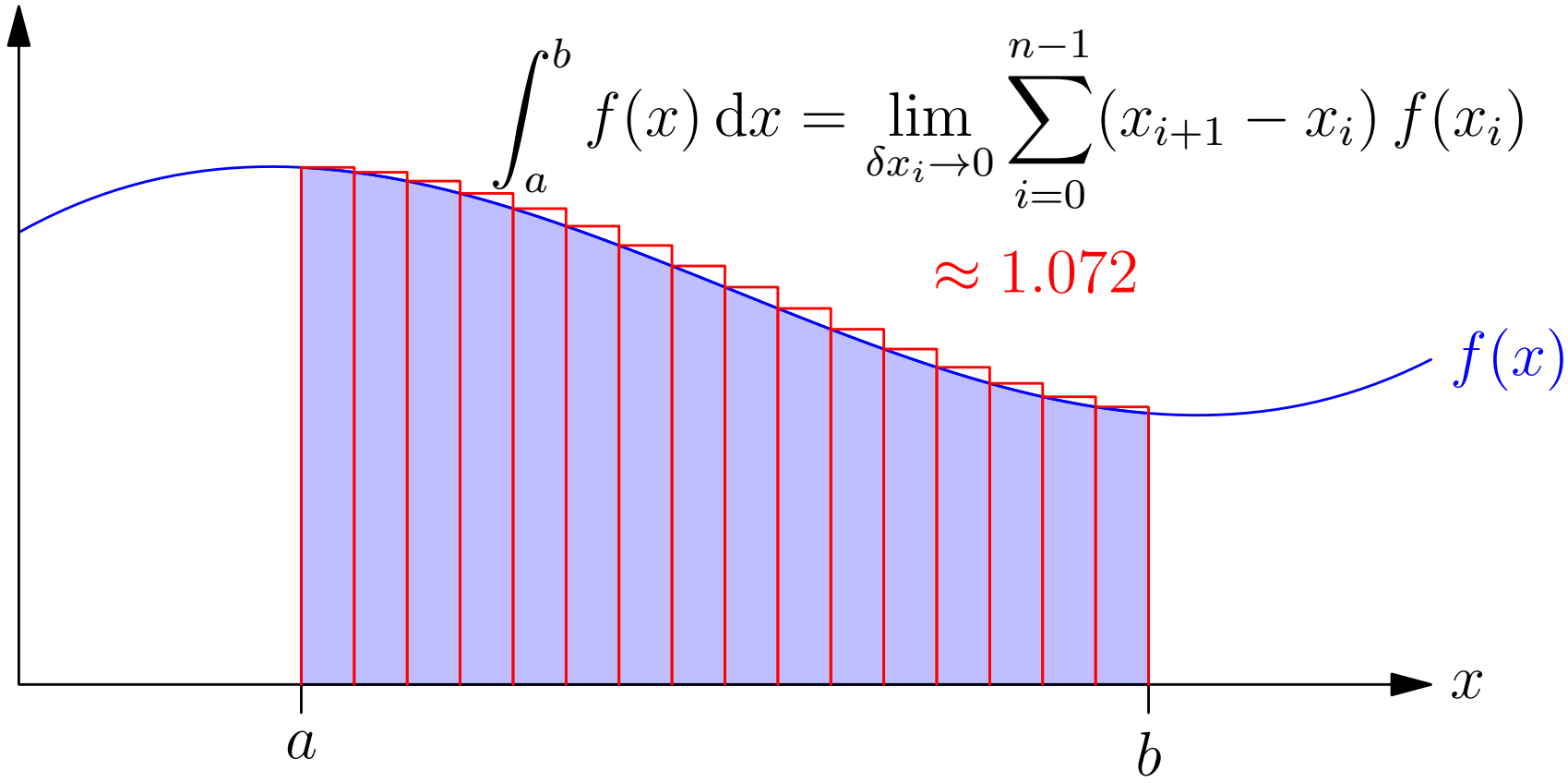
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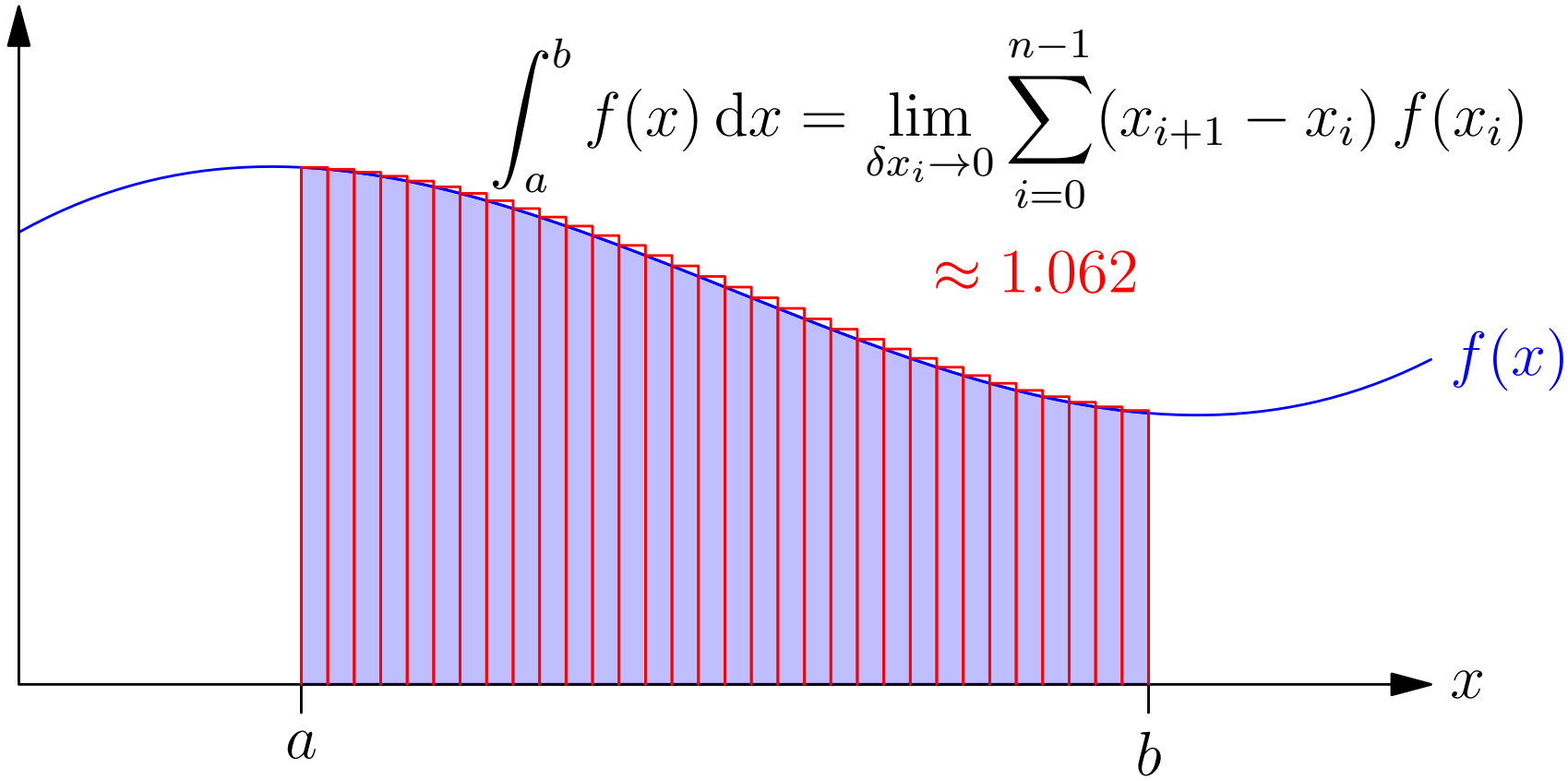
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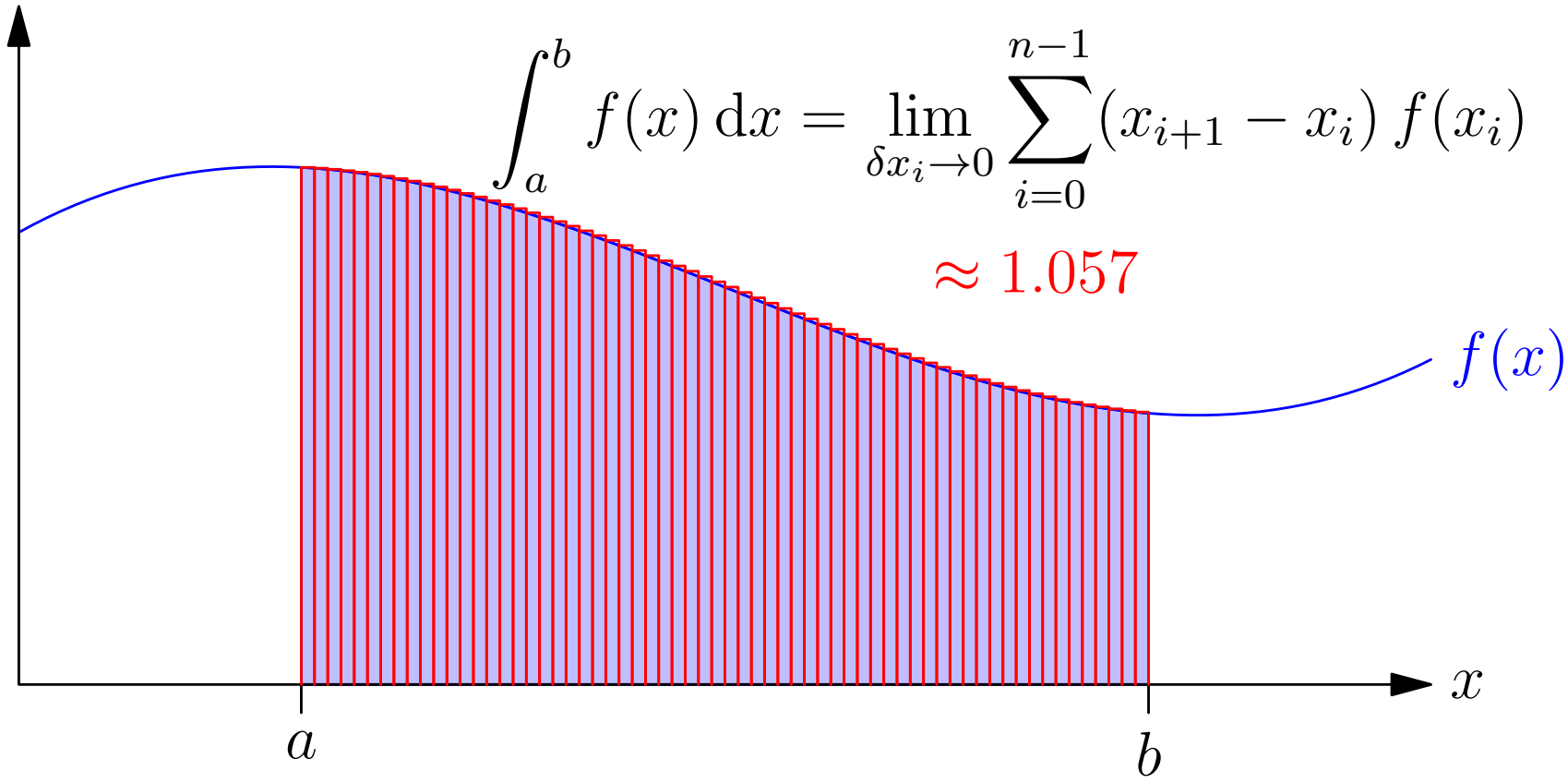
Riemann Integral

- Integrals represent area beneath a curve



Riemann Integral

- Integrals represent area beneath a curve



Linearity of Integration

- Integration is a linear operator

$$\int_a^b (r\,f(x) + s\,g(x))\,\mathrm{d}x = \lim_{\delta x_i \to 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i)(r\,f(x_i) + s\,g(x_i))$$

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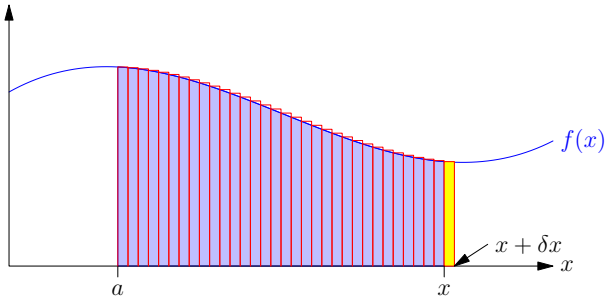
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$$= r \int_a^b f(x)\,\mathrm{d}x + s \int_a^b g(x)\,\mathrm{d}x$$

Fundamental Theorem of Calculus

- Let

$$I(a,x) = \int_a^x f(z)\mathrm{d}z = \lim_{\delta z_i \to 0} \sum_{i=0}^{n-1} (z_{i+1} - z_i) f(z_i)$$



- Now for small δx

$$I(a,x+\delta x) = \int_a^{x+\delta x} f(z)\mathrm{d}z$$

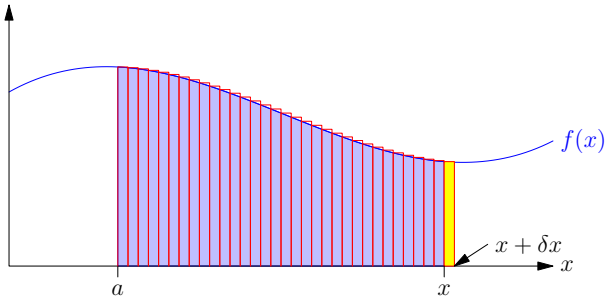
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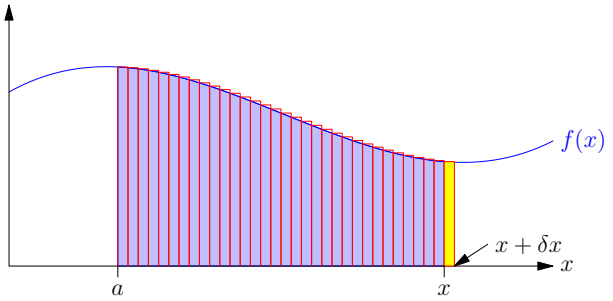
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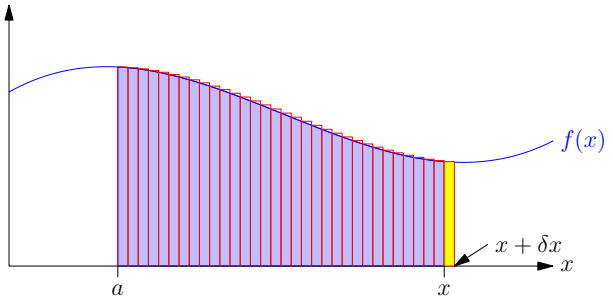
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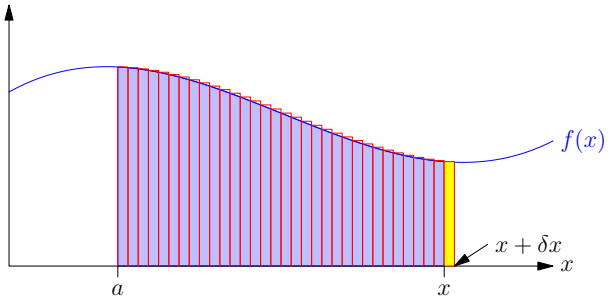
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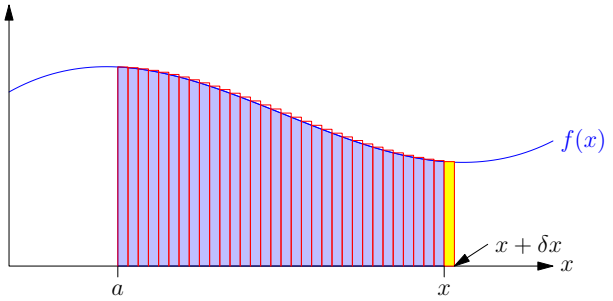
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The Other Way Around

- Consider

$$\int_a^b \frac{\mathrm{d}f(x)}{\mathrm{d}x}\mathrm{d}x = \int_a^b \lim_{\delta x \to 0} \frac{f(x+\delta x) - f(x)}{\delta x}\mathrm{d}x$$

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$$\begin{aligned}
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&= \lim_{x_{i+1}-x_i \to 0} \sum_{i=0}^{n-1} (x_{i+1} - x_i) \frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} \\
&= \lim_{x_{i+1}-x_i \to 0} \sum_{i=0}^{n-1} (f(x_{i+1}) - f(x_i)) \\
&= (f(x_1) - f(x_0)) + (f(x_2) - f(x_1)) + (f(x_3) - f(x_2)) + \cdots \\
&\quad + (f(x_{n-1}) - f(x_{n-2})) + (f(x_n) - f(x_{n-1})) \\
&= f(x_n) - f(x_0) = f(b) - f(a)
\end{aligned}$$

- We can think of integration as an **anti-derivative**—it undoes differentiation

Indefinite Integrals

- So far we have considered **definite integrals** where we integrate between two points (a and b)
- However, when think about integration as an anti-derivative, it is useful to think of a function $F(x) = \int f(x)\mathrm{d}x$
- So that $F'(x) = f(x)$
- However the function $F(x)$, $F(x) + 1$, $F(x) + \pi$, etc. all have the same derivative so $F(x)$ is only defined up to an additive constant
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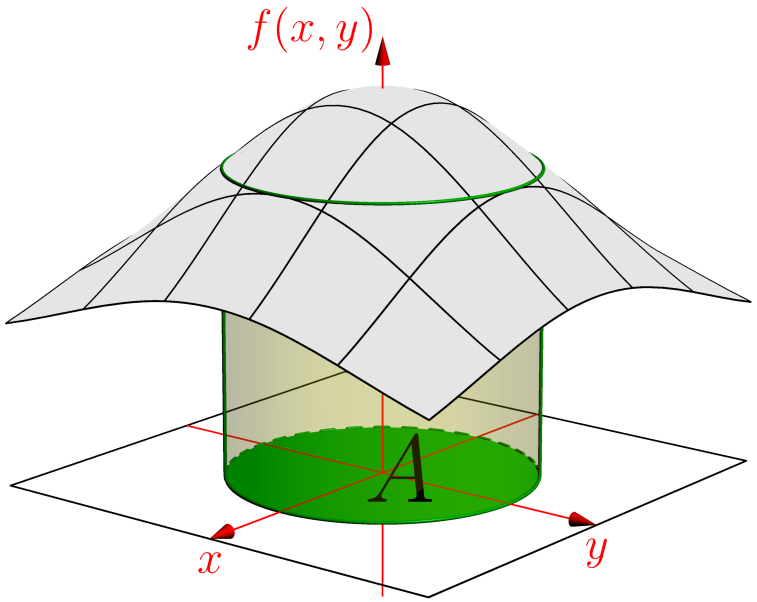
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Multiple Integrals

- For functions involving many independent variables (e.g. $f(x,y)$, $f(x,y,z)$, $f(\boldsymbol{x})$) we can integrate over multiple dimensions

- For example

$$\iint_A f(x,y)\,\mathrm{d}x\,\mathrm{d}y$$



- It gets tedious writing multiple integral signs and I tend to write just one

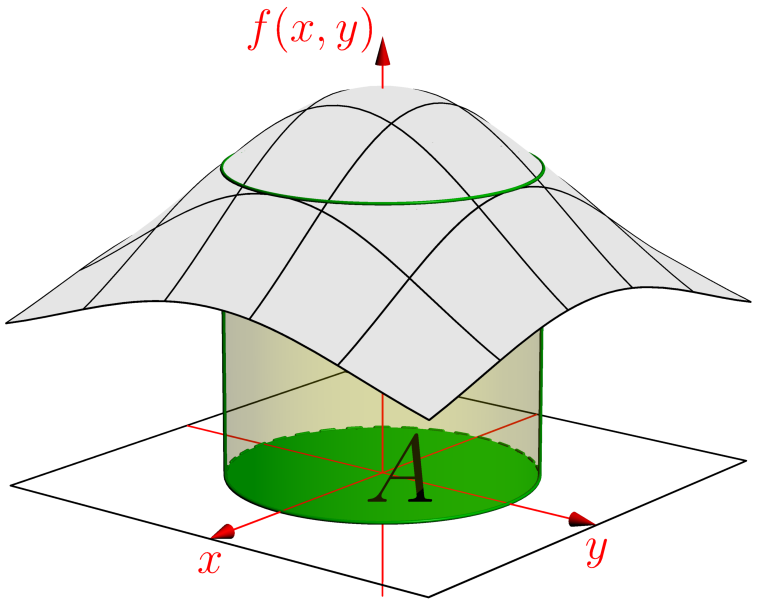
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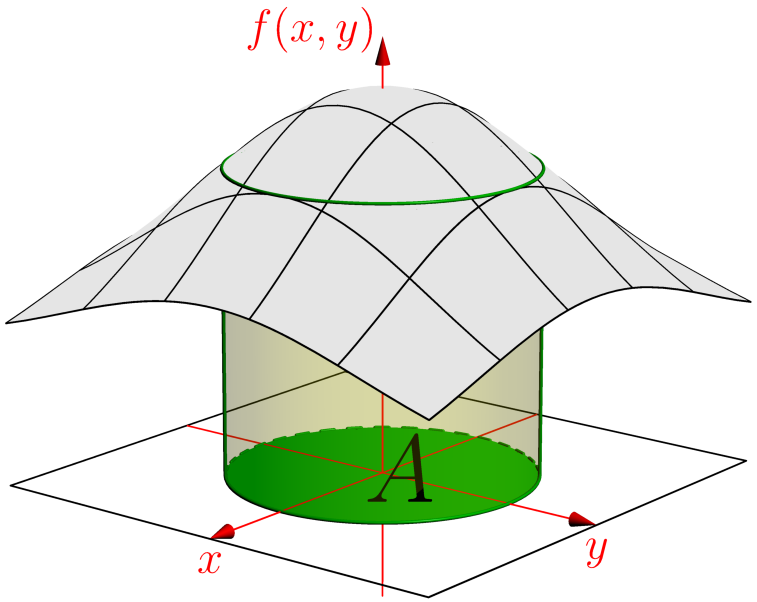
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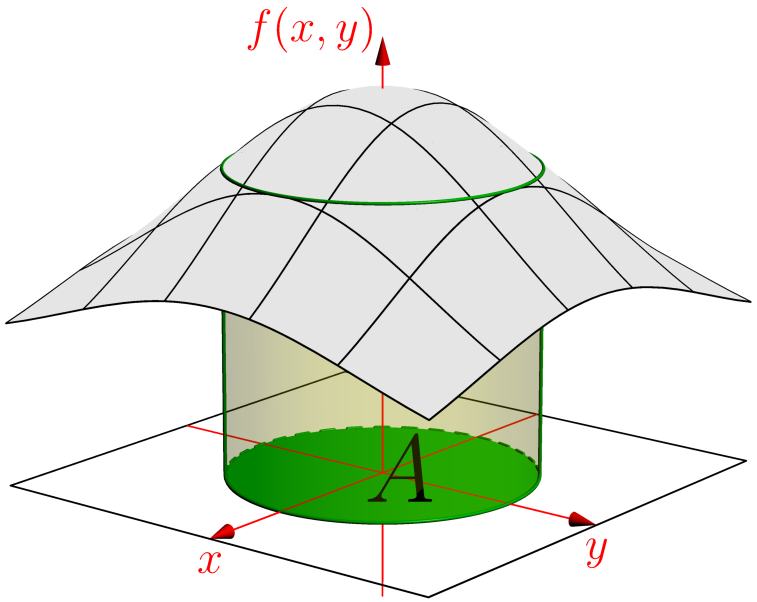
$$\int\cdots\int f(x_1,x_2,\ldots,x_n)\mathrm{d}x_1\mathrm{d}x_2\cdots\mathrm{d}x_n$$

Multiple Integrals

- For functions involving many independent variables (e.g. $f(x,y)$, $f(x,y,z)$, $f(\boldsymbol{x})$) we can integrate over multiple dimensions

- For example

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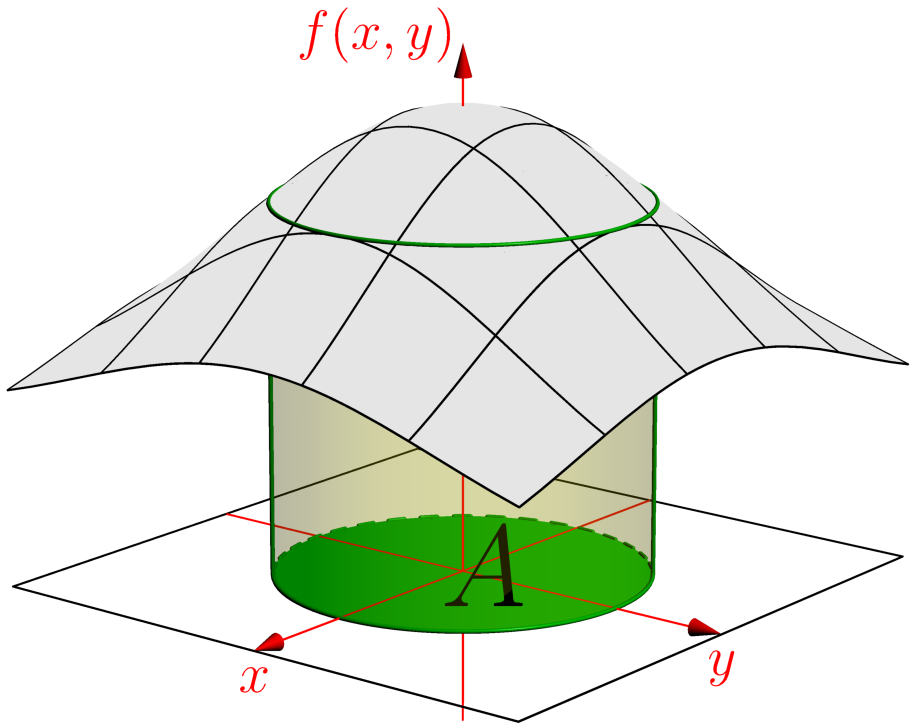


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$$\int \cdots \int f(x_1,x_2,\ldots,x_n)\mathrm{d}x_1\mathrm{d}x_2\cdots\mathrm{d}x_n = \int f(\boldsymbol{x})\mathrm{d}\boldsymbol{x}$$

Outline

1. Defining Integrals
2. **Doing Integrals**
3. Changing Variables



Performing Integration

- A key method for performing integrals is through knowledge of the anti-derivative
- If we know $F'(x) = f(x)$ then $F(x) + c = \int f(x)\mathrm{d}x$
- E.g. we know that $\mathrm{d}x^n/\mathrm{d}x = nx^{n-1}$ therefore

$$\int x^{n-1}\mathrm{d}x = \frac{1}{n}\int \frac{\mathrm{d}x^n}{\mathrm{d}x}\mathrm{d}x$$

and

$$\int_a^b x^{n-1}\mathrm{d}x = \frac{b^n}{n} - \frac{a^n}{n}$$

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Is Integration Straightforward?

- We saw, due to the product and chain rules, that we can differentiate almost anything

- Products and compositions

$$\int f(x)g(x)\mathrm{d}x = ? \qquad \int f(g(x))\mathrm{d}x = ?$$

- Unfortunately, unlike differentiation we don't have a small parameter we can expand in

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Integration by Parts

- Recall the product rule $\frac{\mathrm{d}f(x)g(x)}{\mathrm{d}x} = \frac{\mathrm{d}f(x)}{\mathrm{d}x}g(x) + f(x)\frac{\mathrm{d}g(x)}{\mathrm{d}x}$
- Integrating we get

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whether this is helpful depends on $f(x)$ and $g(x)$

Example of Integration by Parts

- Consider

$$\Pi(z) = \int_0^\infty x^z \mathrm{e}^{-x} \mathrm{d}x$$

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- Now
$\Pi(n) = n\Pi(n-1) = n(n-1)\Pi(n-2) = n(n-1)(n-2)\ldots 1 = n!$

Differentiating Through the Integral

- A trick that sometimes works is differentiating through an integral, e.g. consider finding moments

$$M_n = \mathbb{E}[X^n] = \int_{-\infty}^{\infty} x^n f_X(x) \mathrm{d}x$$

- We can define a moment generating function

$$Z(\ell) = \int_{-\infty}^{\infty} \mathrm{e}^{\ell x} f_X(x) \mathrm{d}x$$

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Cumulant Generating Function

- Note that $\mathrm{e}^{\ell x} = 1 + \ell x + \frac{1}{2}\ell^2 x^2 + \frac{1}{3!}\ell^3 x^3 + \cdots$
- So

$$Z(\ell) = \int_{-\infty}^{\infty} \mathrm{e}^{\ell x} f_X(x) \mathrm{d}x$$

- Now using $\log(1+\epsilon) = \epsilon - \frac{1}{2}\epsilon^2 + \frac{1}{3}\epsilon^3 + \cdots$

$$G(\ell) = \log(Z(\ell)) = \ell M_1 + \frac{1}{2}\ell^2\left(M_2 - M_1^2\right) + \frac{1}{3!}\ell^3\left(M_3 - 3M_2M_1 + 2M_1^3\right) + \cdots$$

- So that $\kappa_n = G^{(n)}(0)$, with $\kappa_1 = M_1$ (the mean), $\kappa_2 = M_2 - M_1^2$ (the variance), $\kappa_3 = M_3 - 3M_2M_1 + 2M_1^3$ (the third cumulant related to the skewness)

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- Note that $\mathrm{e}^{\ell x} = 1 + \ell x + \frac{1}{2}\ell^2 x^2 + \frac{1}{3!}\ell^3 x^3 + \cdots$
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- Surprisingly integration sometimes is easier when carried out in the complex plane
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- Interestingly, also there is an algorithm that allows us to integrate a lot of function. It is sufficiently complicated that you need to write a computer algorithm of considerable complexity to implement it. Most symbolic manipulation packages (e.g. Mathematica) have implemented some part of this algorithm

Special Functions

- There are integrals with no known closed form solution
- We saw that $\Pi(z) = \int_0^\infty x^z \mathrm{e}^{-x} \mathrm{d}x$ satisfies $\Pi(z) = z\Pi(z-1)$
- For integer n then $\Pi(n) = n!$, but for general z, the integal $\Pi(z)$ can't be written in terms of elementary functions
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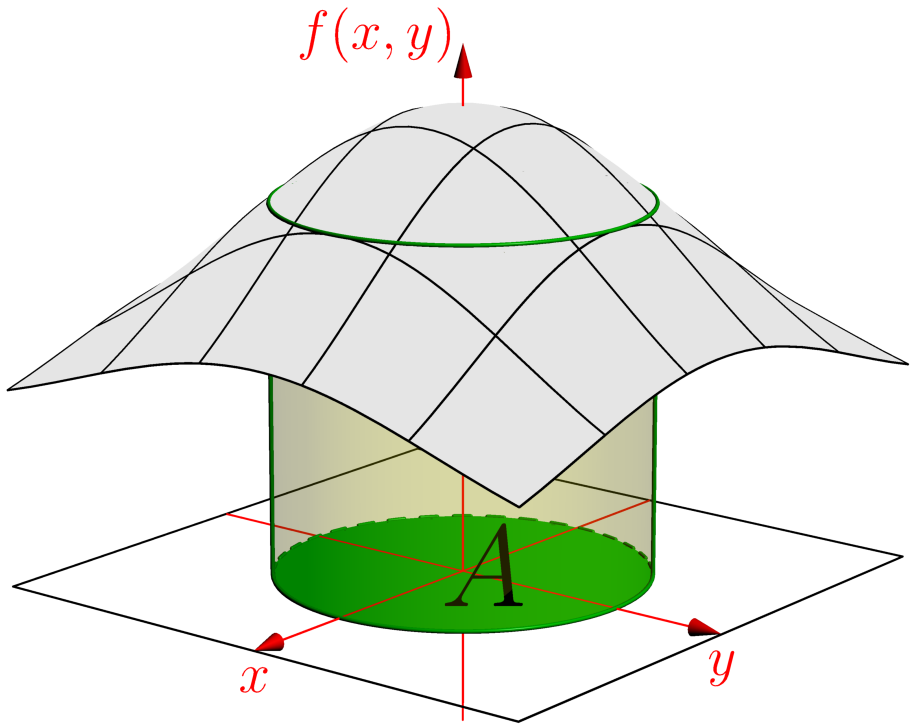
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- Other special function defined by integrals exist (e.g. the Bessel , Aire, hypergeometric, elliptic, error functions, . . .)

Outline

1. Defining Integrals
2. Doing Integrals
3. **Changing Variables**



Substitution

- We can make a transformation from x to u

$$\int_a^b f(x)\mathrm{d}x = \lim_{\delta x_i \to 0} \sum_{i=0}^{n-1} f(x_i)(x_{i+1} - x_i)$$

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  - ⋆ where u_i is such that $x(u_i) = x_i$ or $u_i = u(x_i)$ where $u(x)$ is the inverse of $x(u)$

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$$= \int_{u(a)}^{u(b)} f(x(u)) \frac{\mathrm{d}x(u)}{\mathrm{d}u} \mathrm{d}u$$

  - ⋆ where u_i is such that $x(u_i) = x_i$ or $u_i = u(x_i)$ where $u(x)$ is the inverse of $x(u)$
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Substitution in Practice

- A recipe
  1. Choose $u(x)$: often an "inner" function whose derivative also appears in the integrand
  2. Write $\mathrm{d}u = u'(x)\mathrm{d}x$ and replace every x by u
  3. Change the limits: $x = a$ becomes $u = u(a)$
  4. Integrate in u

- Example: $\displaystyle\int_0^1 2x(1+x^2)^3\mathrm{d}x$

$$u = 1 + x^2, \quad \mathrm{d}u = 2x\mathrm{d}x, \quad x : 0 \to 1 \;\Rightarrow\; u : 1 \to 2$$

$$\int_0^1 2x(1+x^2)^3\mathrm{d}x = \int_1^2 u^3\mathrm{d}u = \left[\frac{u^4}{4}\right]_1^2 = \frac{16-1}{4} = \frac{15}{4}$$

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- The limits can swap over: for $\int_0^{\pi/2} \sin(x)\cos^2(x)\mathrm{d}x$ take $u = \cos(x)$, $\mathrm{d}u = -\sin(x)\mathrm{d}x$, and $x : 0 \to \pi/2$ gives $u : 1 \to 0$

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- The factor σ is the one-dimensional version of the Jacobian determinant (next slides)

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Gamma Distribution

- Last lecture we met the Gamma distribution

$$\text{Gamma}(x|a,b) = cx^{a-1}\mathrm{e}^{-bx}, \qquad x \geq 0$$

  Let's compute the normalisation factor c

- We need $c\int_0^\infty x^{a-1}\mathrm{e}^{-bx}\mathrm{d}x = 1$

- Substitute $z = bx$, so $\mathrm{d}x = \mathrm{d}z/b$ and the limits stay 0 and ∞

$$c\int_0^\infty \left(\frac{z}{b}\right)^{a-1}\mathrm{e}^{-z}\frac{\mathrm{d}z}{b} = \frac{c}{b^a}\int_0^\infty z^{a-1}\mathrm{e}^{-z}\mathrm{d}z$$

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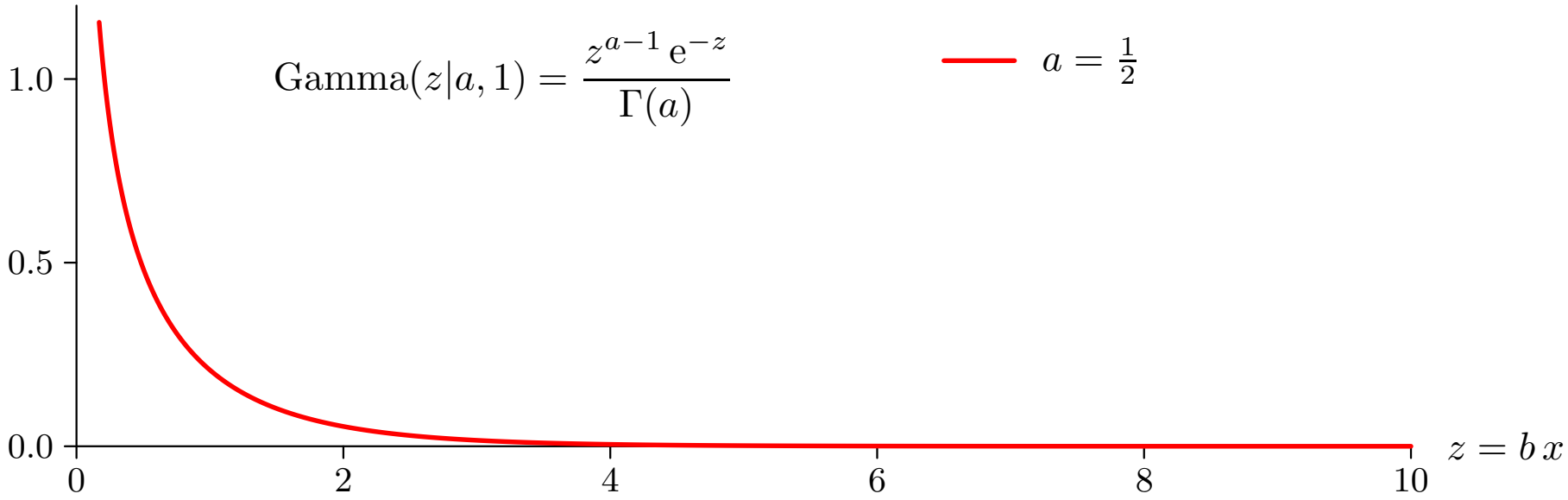
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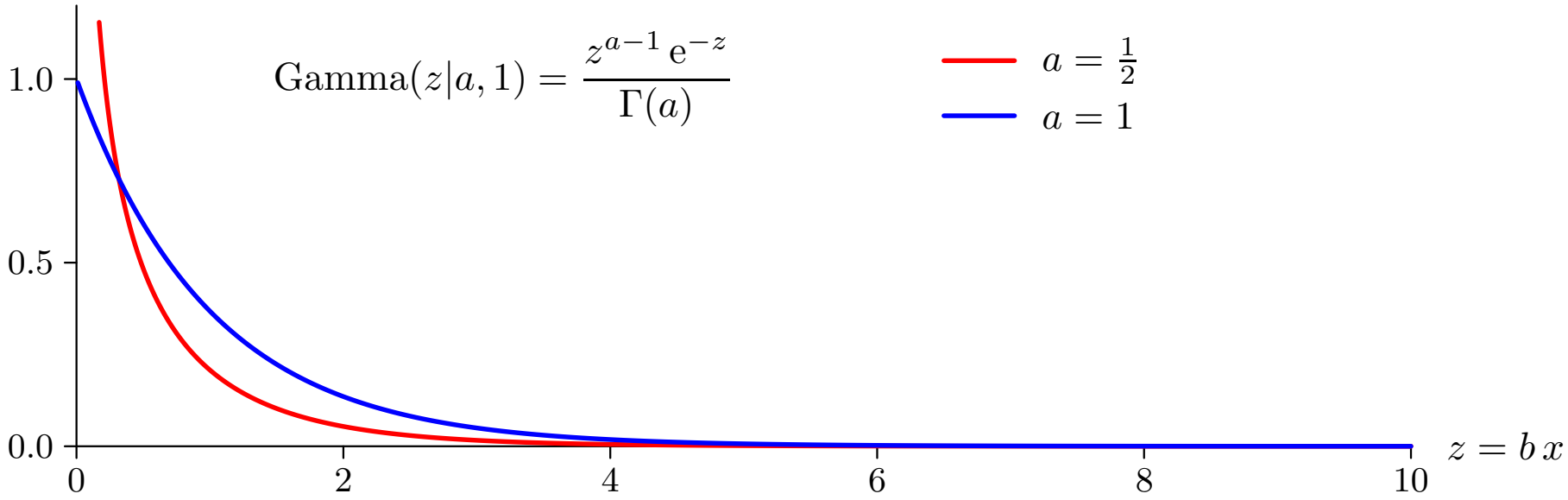
- So $c = \dfrac{b^a}{\Gamma(a)} = \dfrac{b^a}{\Pi(a-1)}$: b only sets the scale, a sets the shape

Shapes of the Gamma Distribution



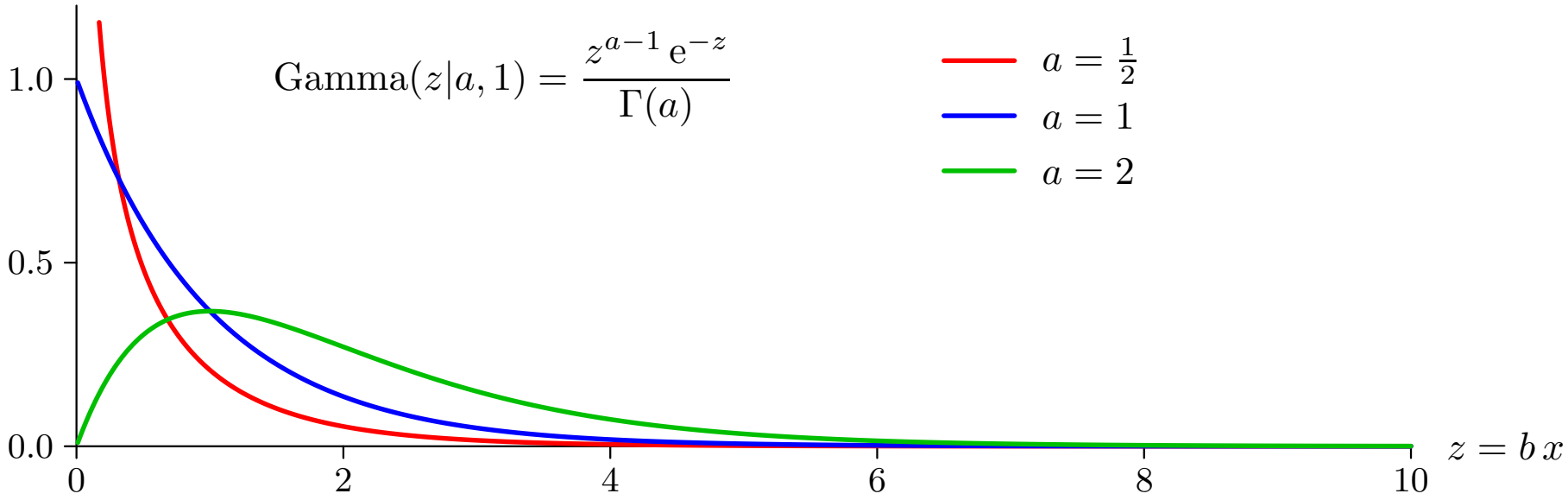
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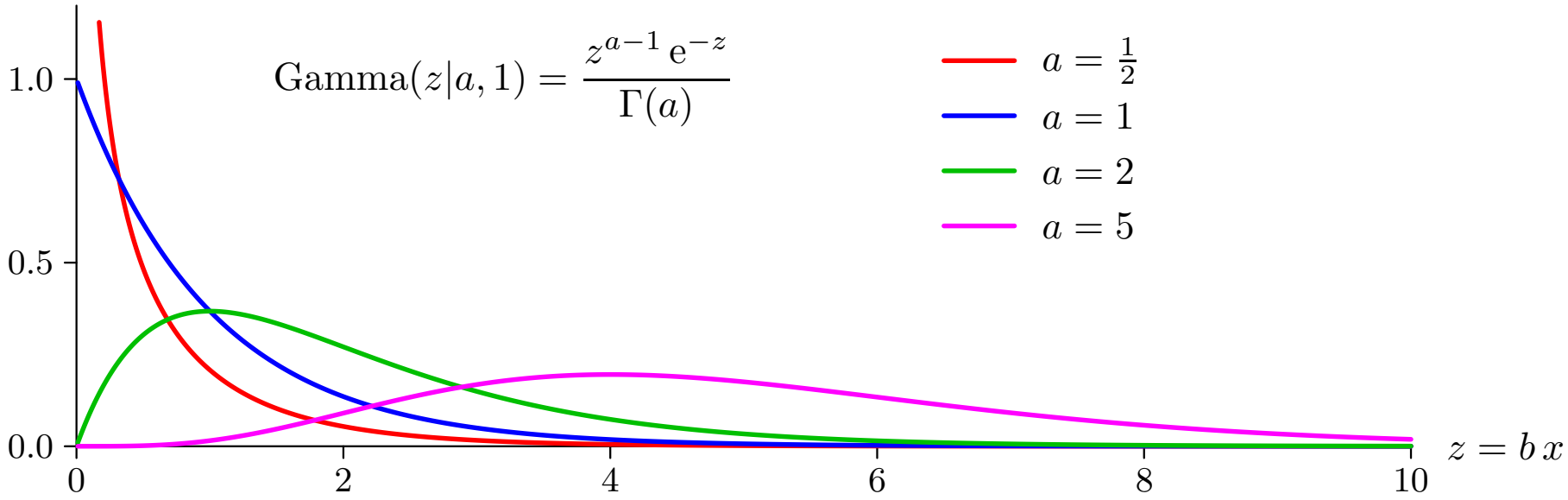
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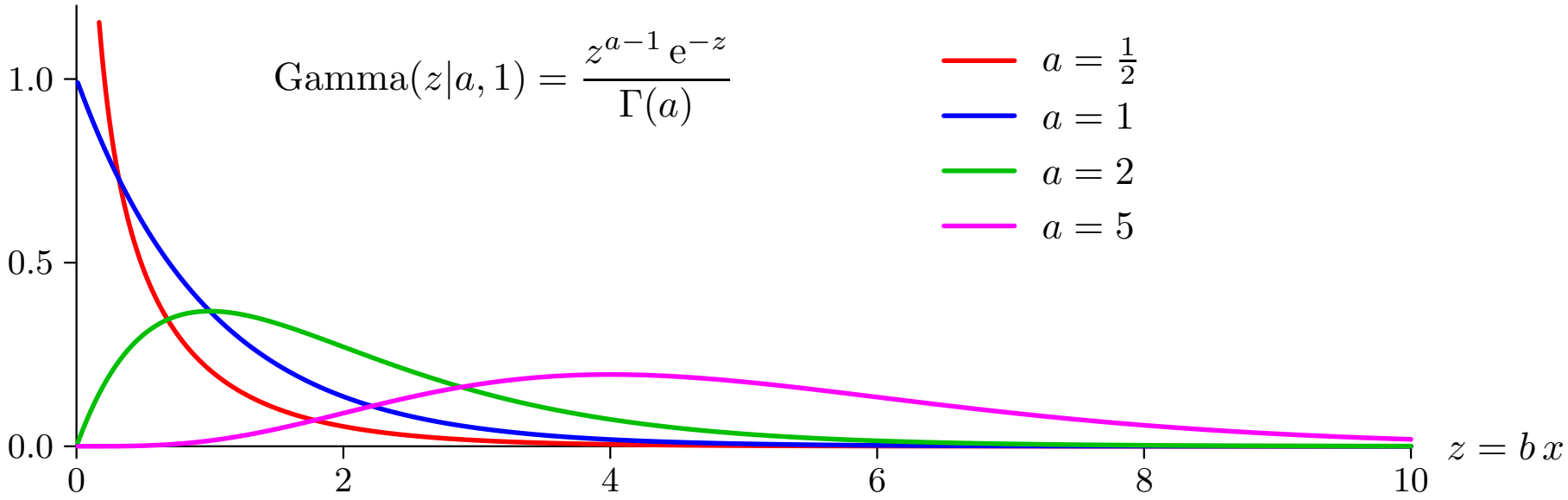
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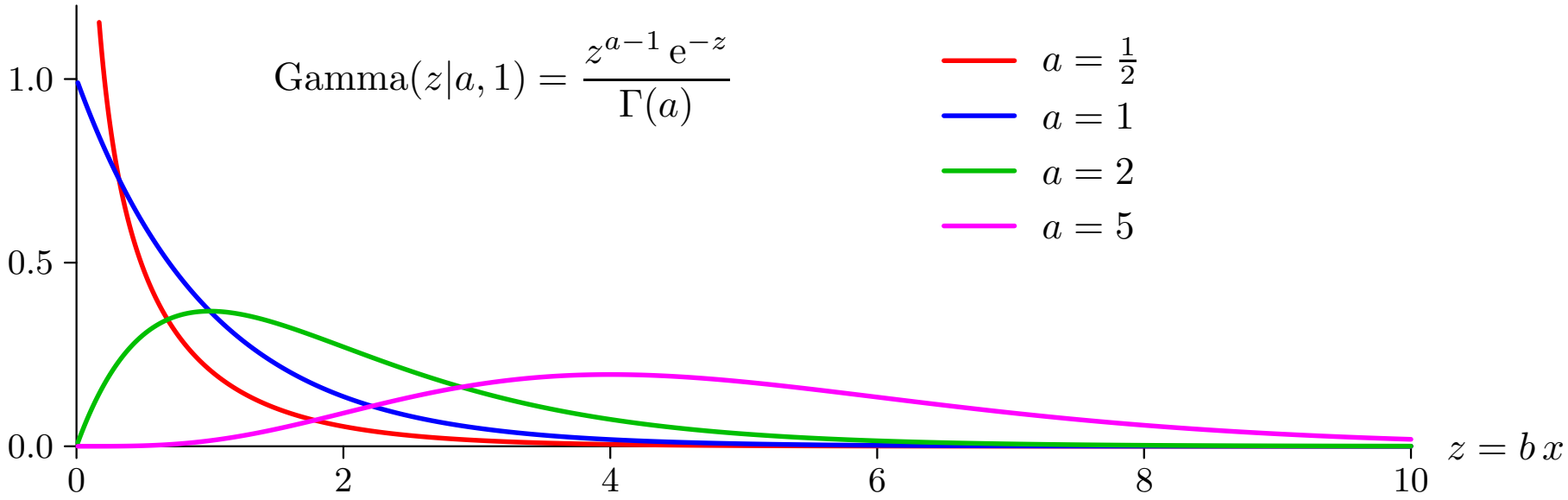
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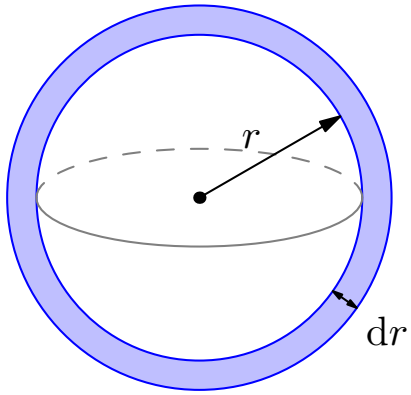
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Lengths of Gaussian Vectors

- Take $\boldsymbol{x} \in \mathbb{R}^n$ with $\boldsymbol{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$: the density $\propto \mathrm{e}^{-r^2/2}$ depends only on $r = \|\boldsymbol{x}\|$



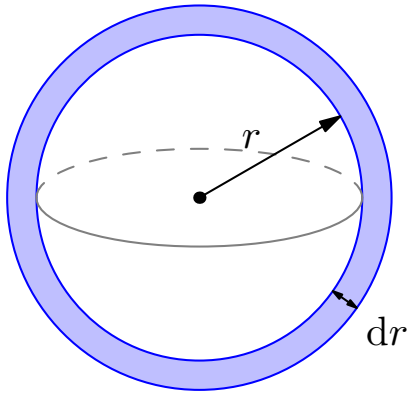
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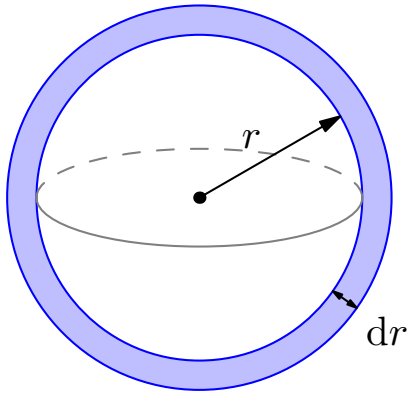
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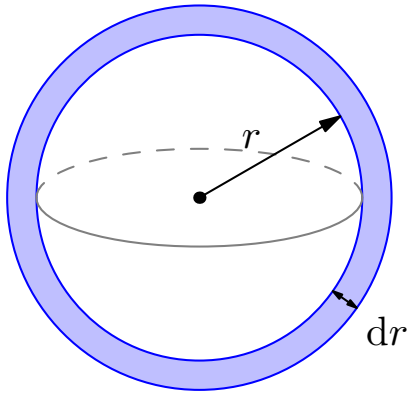
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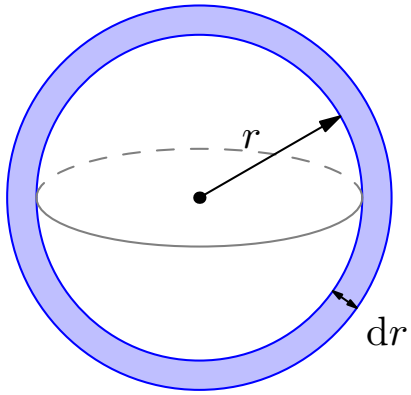
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Gamma Distributions Everywhere

- **Chi-squared**: a sum of squares of n independent standard normals is $\mathrm{Gamma}(\frac{n}{2},\frac{1}{2})$
- For m normal samples the sample variance has $(m-1)\hat{\sigma}^2/\sigma^2 \sim \chi^2_{m-1}$, which lies behind the t-test
- **Maxwell–Boltzmann**: each velocity component of a gas molecule of mass M is $\mathcal{N}(0,kT/M)$, so the speed is the $n=3$ case, $f(v) \propto v^2 \mathrm{e}^{-Mv^2/(2kT)}$, and the energy $\frac{1}{2}Mv^2 \sim \mathrm{Gamma}(\frac{3}{2},\frac{1}{kT})$
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Changing Variables in Multidimensional Space

- When changing variables in many dimensions $\boldsymbol{x} \rightarrow \boldsymbol{u}$ the change of variables involves the Jacobian

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$$\int f(\boldsymbol{x})\mathrm{d}\boldsymbol{x} = \int f(\boldsymbol{x}(\boldsymbol{u}))|\det(\mathbf{J})|\mathrm{d}\boldsymbol{u}, \qquad \boldsymbol{J} = \begin{pmatrix} \frac{\partial x_1}{\partial u_1} & \frac{\partial x_1}{\partial u_2} & \cdots & \frac{\partial x_1}{\partial u_n} \\ \frac{\partial x_2}{\partial u_1} & \frac{\partial x_2}{\partial u_2} & \cdots & \frac{\partial x_2}{\partial u_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial u_1} & \frac{\partial x_n}{\partial u_2} & \cdots & \frac{\partial x_n}{\partial u_n} \end{pmatrix}$$

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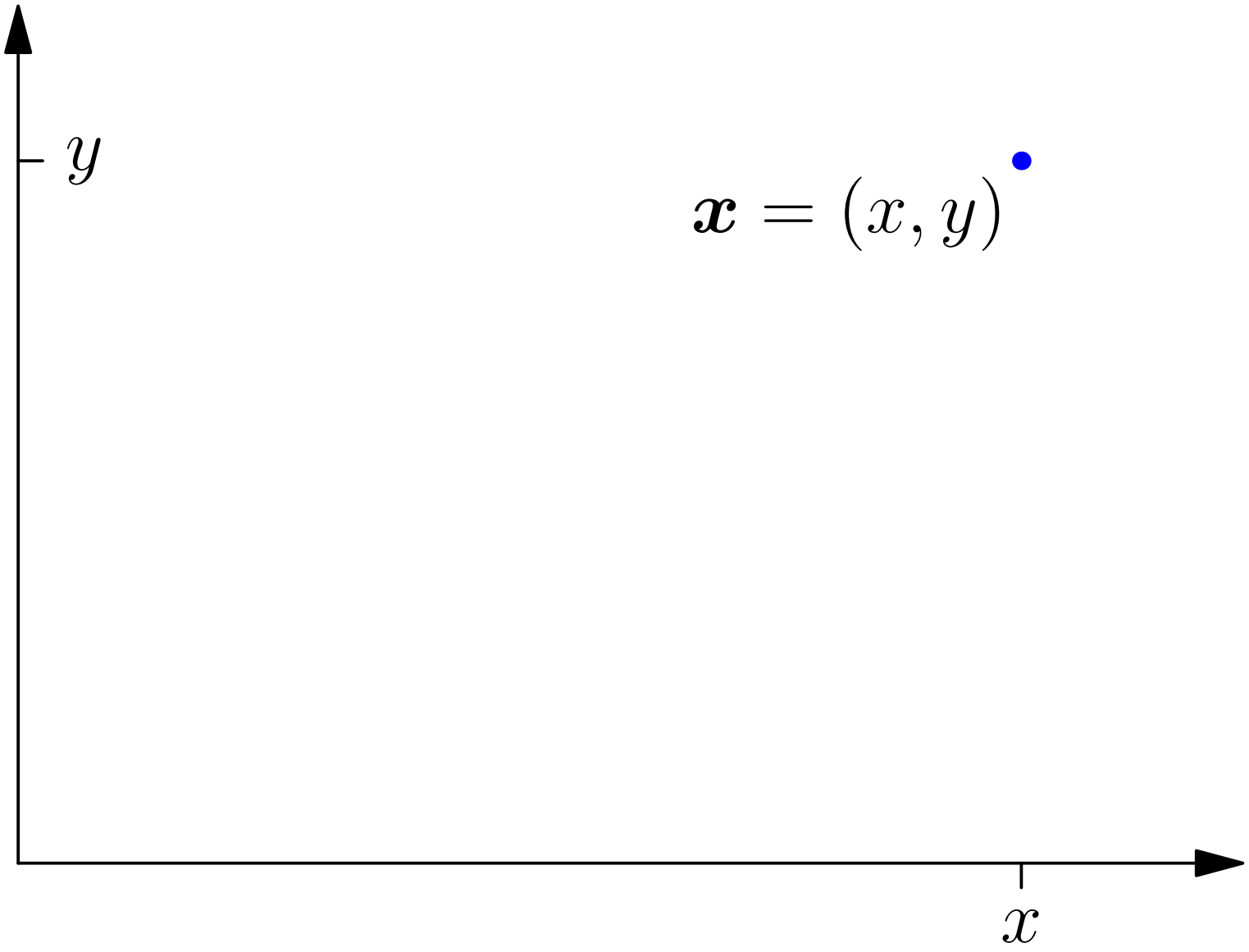
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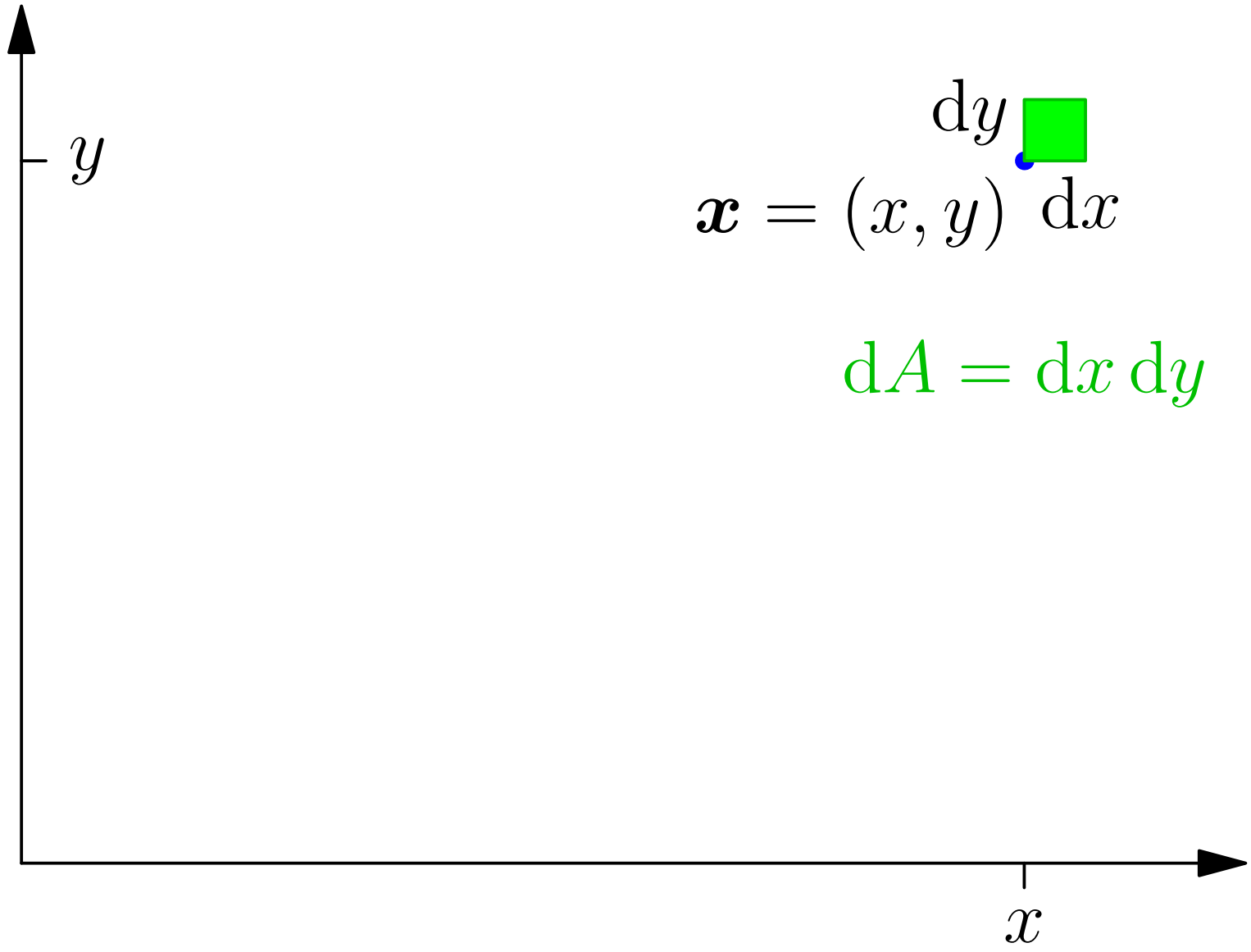
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- That is, $\mathrm{d}x\mathrm{d}y = r\mathrm{d}r\mathrm{d}\theta$

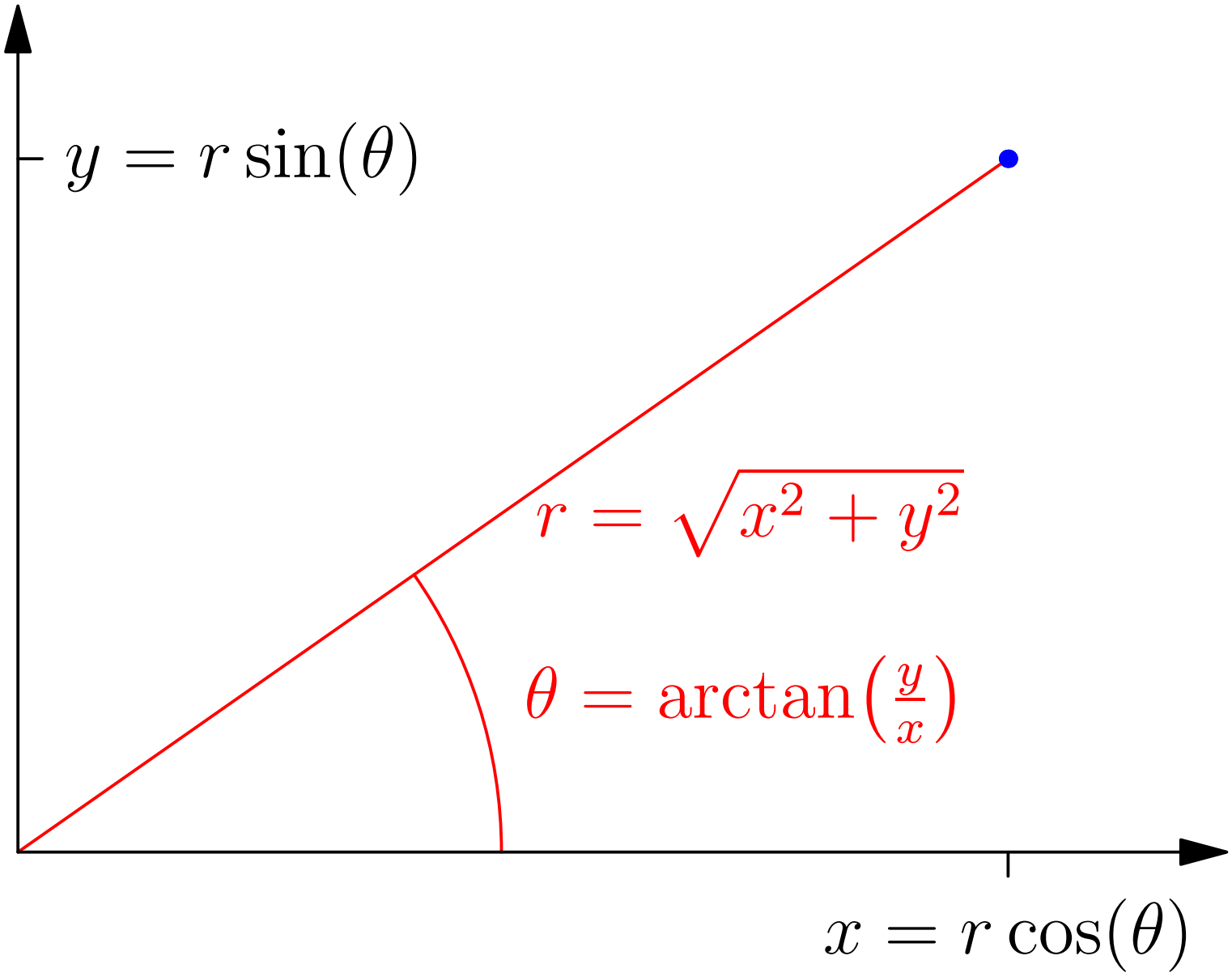
Change of Variables in Pictures



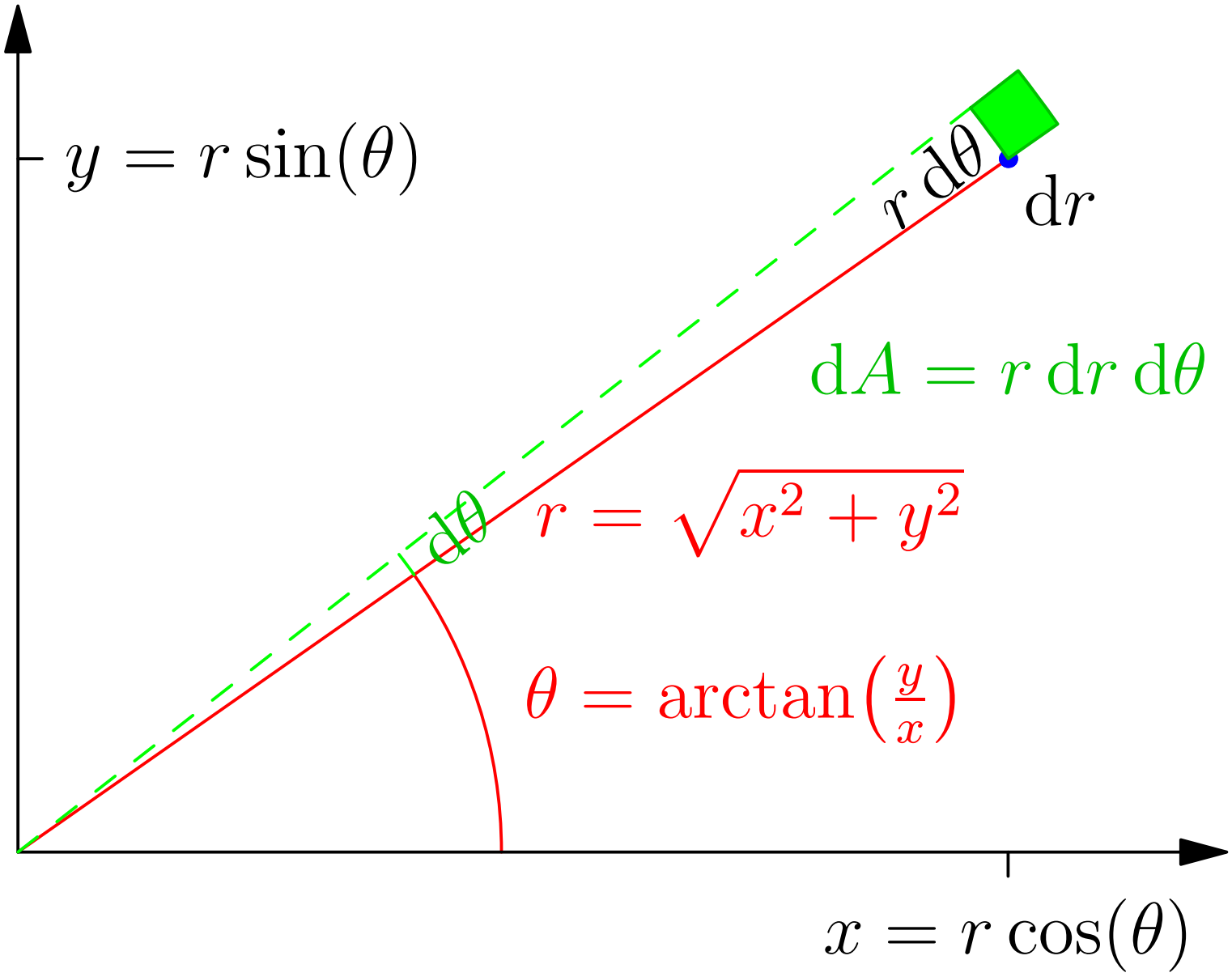
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- Making friends with integration will give you a super-power that not too many people share